

ECE 647: Lecture 11 Composition of Functions

1 Differentiation

Starting with the function $f(x)$ defined as composition of two functions such that:

$$f(x) = h(g(x))$$

Here, the functions h and g are termed the *outer* and *inner* functions, respectively.

Taking the first derivative of $f(x)$ by applying the chain rule yields:

$$f'(x) = h'(g(x)) \cdot g'(x)$$

Taking the second derivative (differentiating $f'(x)$) yields:

$$f''(x) = h''(g(x)) \cdot \underbrace{(g'(x))^2}_{\geq 0} + h'(g(x)) \cdot g''(x)$$

2 Convexity

Given a function $f(x)$ defined as a composition of h and g the following rules apply:

- Assume g is convex, i.e., $g''(x) \geq 0$
 - If h is convex, h is non-decreasing:

$$h''(x) \geq 0, \quad h'(x) \geq 0$$

then f is **convex**

- If h is concave, h is non-increasing:

$$h''(x) \leq 0, \quad h'(x) \leq 0$$

then f is **concave**

- Assume g is concave, i.e., $g''(x) \leq 0$:

- If h is concave, h is non-decreasing:

$$h''(x) \leq 0, \quad h'(x) \geq 0$$

$\Rightarrow f$ is concave

- If h is convex, h is non-increasing:

$$h''(x) \geq 0, \quad h'(x) \leq 0$$

$\Rightarrow f$ is convex

Note

- The convexity or concavity of f must follow that of h . This means that if h is convex, f should also be convex under the given conditions, and if h is concave, then f should be concave.
- If h and g are of the same type (both convex or both concave), then for f to inherit the same property, h must be non-decreasing. That is, its first derivative $h'(x)$ must satisfy $h'(x) \geq 0$.
- If h and g are of different types (one is convex while the other is concave), then for f to maintain the correct convexity or concavity, h must be non-increasing. This means its first derivative $h'(x)$ should satisfy $h'(x) \leq 0$.

Examples

- (a) If g is convex, then $e^{g(x)}$ is convex.

$$h(x) = e^x$$

(b) If g is convex, then $g^2(x)$ may not be convex.

$$h(x) = x^2$$

(c) If g is convex, and $g(x) \geq 0$, then $g^3(x)$ is convex.

If $g : \mathbb{R}^n \rightarrow \mathbb{R}$, take a restriction to a line:

$$f(\lambda) = h(g(x + \lambda v))$$

If g is convex/concave in x , then:

$$g(x + \lambda v) \text{ is convex/concave in } \lambda$$

$$g(x) \text{ is convex/concave in } x \iff g(x + \lambda v) \text{ is convex/concave in } \lambda$$

dom(f) $\neq \mathbb{R}$ When $\text{dom}(f) \neq \mathbb{R}$ special consideration needs to be applied. Below is an example:

$$g(x) = x^2$$

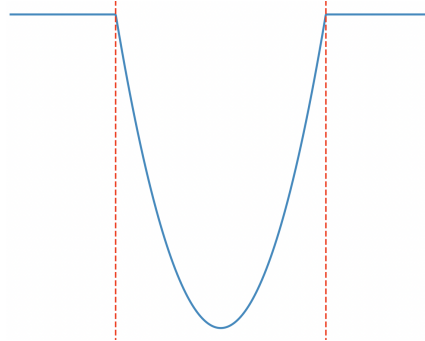
$$h(x) = 0 \quad \text{with } \text{dom}(h) = [1, 2]$$

$$f(x) = h(x^2)$$

- h is convex.
- h is non-decreasing.
- g is convex.

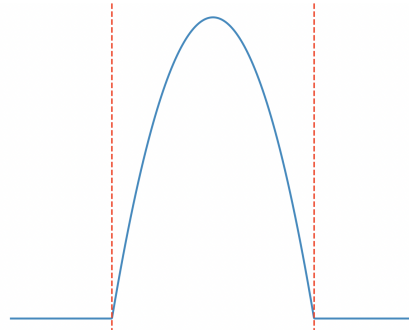
One of the safest approaches is to create an **extended value function**. Given a $\text{dom}(f) \not\subset \mathbb{R}$, create $\tilde{\mathbb{R}}$. Then we have the following: If h is convex over its domain:

$$\tilde{h}(x) = \begin{cases} h(x) & \text{if } x \in \text{dom}(h) \\ +\infty & \text{if } x \notin \text{dom}(h) \end{cases}$$



If h is concave over its domain:

$$\tilde{h}(x) = \begin{cases} h(x) & \text{if } x \in \text{dom}(h) \\ -\infty & \text{if } x \notin \text{dom}(h) \end{cases}$$



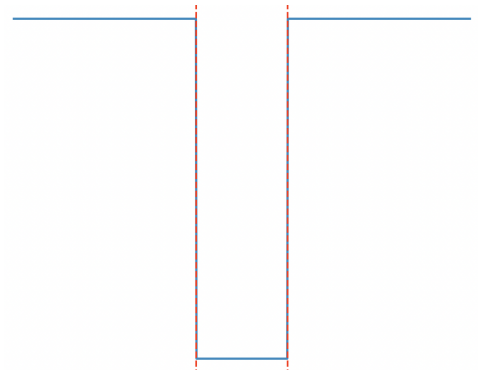
Rule: If $\text{dom}(h)$ is not $\mathbb{R}$, then to check convexity of $h(g(x))$, we need to use properties of $\tilde{h}$.

Note that:

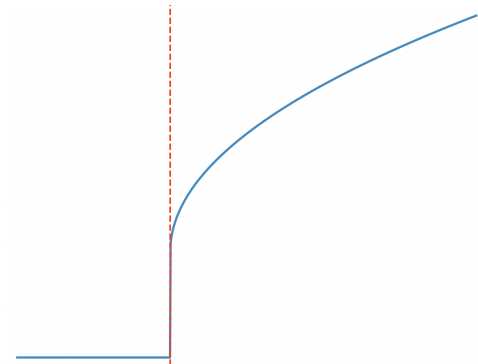
- $\tilde{h}$ always preserves convexity/concavity of h .
- $\tilde{h}$ always preserves non-decreasing/non-increasing properties of h .
- $\tilde{h}$ is not necessarily non-decreasing or non-increasing as a function of $g(x)$.

Examples:

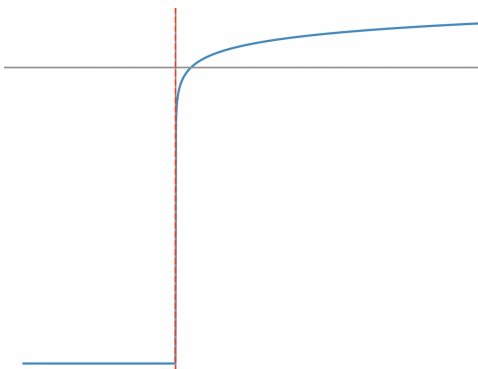
a $h(x) = [1, 2]$



b $h(x) = \sqrt{x}$



c $h(x) = \log(x)$



Example

1. If $g(x)$ is concave, is $\log[g(x)]$ concave?

$$h(x) = \log(x)$$

Then on $\{x \mid g(x) > 0\}$,

$$\tilde{h}(x) \text{ is true on } x \in \mathbb{R}$$

2. If $g(x)$ is concave, is $\frac{1}{g(x)}$ convex?

$$h(x) = \frac{1}{x}$$

If $\{x \mid g(x) > 0\}$, then yes.

3. $h(x) = x^{\frac{3}{2}}$ on $[0, \infty)$, and $\tilde{h}$ is **not** non-decreasing or non-increasing.

$$h'(x) = \frac{3}{2}x^{\frac{1}{2}}$$

$$h''(x) = \frac{3}{4}x^{-\frac{1}{2}} = \frac{3}{4\sqrt{x}}$$

4. Is $h(x^2 - 4x)$ convex?

$$h(x) \text{ is convex on } (0, \infty)$$

Given:

$$g(x) = x^2 - 4x$$

$$g''(x) = 2 \text{ (which is convex)}$$

Conclusion: No, because the domain is

$$x \in (-\infty, 0) \cup (4, \infty)$$

5.

$$h(x) = \begin{cases} x^{\frac{3}{2}}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

and $\tilde{h}$ is convex and non-decreasing.

T Then we have $h(x^2 - 4x)$ is convex.